queues) and expiration times, React can yield control back to the browser mid-render, prioritizing urgent tasks like user input over less critical updates.
- *Effect list batching (pre-commit Phase)*—instead of applying changes immediately, React organizes them into a linked effect list, ensuring all updates are atomic and the UI never enters an inconsistent state.
- *Non-interruptible commit (Commit Phase)*—React synchronously applies all updates in the effect list, runs layout effects (useLayoutEffect), and schedules passive effects (useEffect).

By understanding Fiber's role, developers can design UIs that align with React's scheduling model, minimize unnecessary re-renders using techniques like React.memo, useMemo, and useCallback, and build applications that remain highly responsive even under heavy computational load.

Probing question (advanced): "What are the distinctions between the Virtual DOM and Fiber architecture?"

Answer: Although both the Virtual DOM and Fiber Architecture are core to React rendering process, they serve very different purposes. The Virtual DOM provides a simplified abstraction of the UI for detecting changes, while Fiber is React's internal engine for scheduling and applying those changes efficiently.

Table 4.4 highlights their differences and shows how they complement each other to enable performant, responsive updates.

Table 4.4 Distinctions between Virtual DOM and Fiber Architecture

Aspect	Virtual DOM	Fiber Architecture
Definition	A lightweight JavaScript object tree representing the UI.	React's internal reconciliation engine for managing rendering work.
Purpose	Provides an abstract layer for diffing and detecting changes between UI states.	Handles task scheduling, prioritization, and incremental rendering.